



GRADUATION PROJECT REPORT

Project Title:

Smart water management: IOT Metering for Apartments

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The successful completion of any task is made even more satisfying and euphoric when we acknowledge the people who guided and encouraged us along the way. Our project is a result of a careful combination of research and knowledge, and we are pleased to present it to you.

We would like to express our sincere gratitude to our internal guide, [Eng. Abdul-Aziz Alrodhyan](#) from the Department of Electronic Engineering, for his unwavering support, encouragement, and guidance. His cooperation and valuable suggestions were greatly appreciated.

We would also like to thank all other members who were directly or indirectly involved in the completion of this project.

Chapter One

Abstract

The aim of this graduation project is to develop a smart water management system for apartments using Internet of Things (IoT) technology. The proposed system will consist of several components, including water flow sensors, a Wi-Fi capable microcontroller, a GSM module, and a cloud-based web and app dashboard.

The main objective of this project is to provide a more efficient and cost-effective solution for water metering in apartments. By logging data to a web dashboard and sending SMS notifications to both property owners and tenants regarding the percentage of bill cut, the system will promote transparency and accountability in water usage. Additionally, by providing real-time monitoring of water usage, the system will encourage water conservation and help to reduce waste.

The proposed system has several advantages over traditional water metering methods. By using IoT technology, the system can provide real-time data on water usage, allowing property owners and tenants to monitor their usage and make informed decisions about their consumption. The system is also scalable, allowing for easy expansion to accommodate additional apartments or buildings.

Overall, the proposed IoT-based water metering system offers a promising solution for improving water management in apartments. By providing real-time monitoring, accurate billing, and promoting water conservation, the system has the potential to deliver significant benefits to both property owners and tenants.

Project Introduction

Water management is a critical issue in many urban areas, where growing populations and increasing demand for resources are putting pressure on existing infrastructure. In apartment buildings, water metering is often a source of conflict between property owners and tenants, as traditional metering methods can be inaccurate, unreliable, and expensive.

To address these challenges, this graduation project aims to develop a smart water management system for apartments using Internet of Things (IoT) technology. The proposed system will consist of several components, including water flow sensors, a Wi-Fi capable microcontroller, a GSM module, and a cloud-based web and app dashboard.

The main objective of this project is to provide a more efficient and cost-effective solution for water metering in apartments. By using IoT technology, the proposed system will provide real-time data on water usage, allowing property owners and tenants to monitor their usage and make informed decisions about their consumption. This will improve the accuracy and transparency of water metering, reducing the potential for disputes and conflicts.

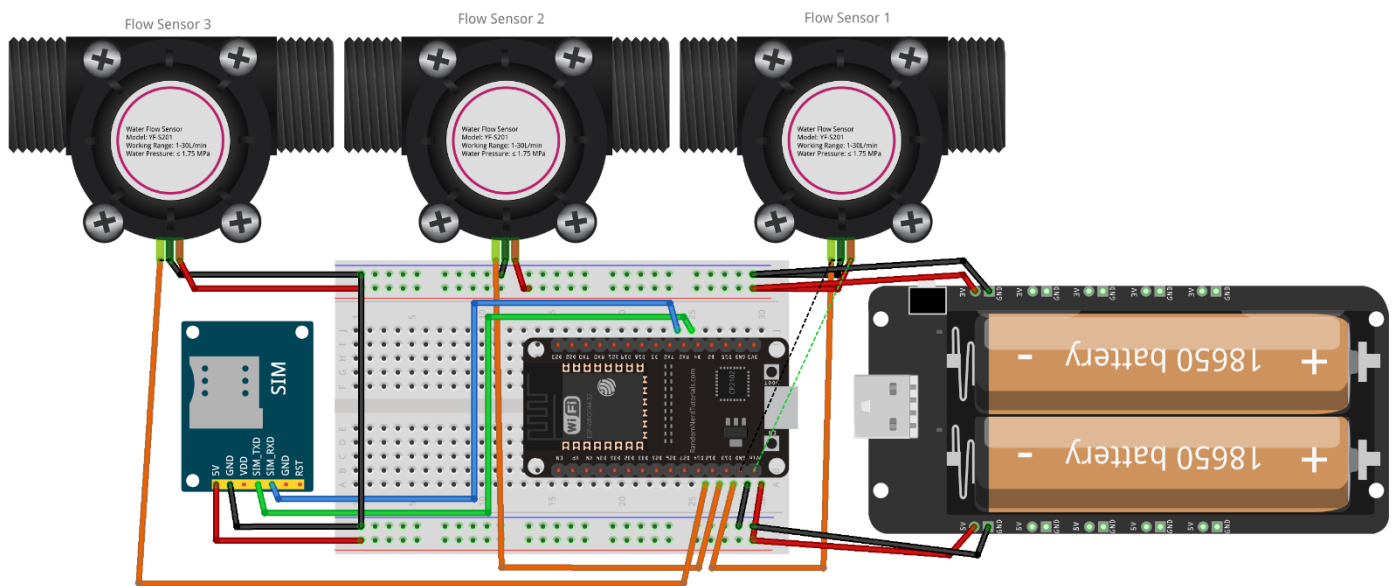
In addition to improving water metering, the proposed system will also promote water conservation. By providing real-time monitoring of water usage, the system will encourage tenants to reduce their consumption and avoid waste. This will not only benefit the environment but also help to reduce costs for both property owners and tenants.

Overall, the proposed IoT-based water metering system offers a promising solution for improving water management in apartments. By providing real-time monitoring, accurate billing, and promoting water conservation, the system has the potential to deliver significant benefits to both property owners and tenants.

Components:

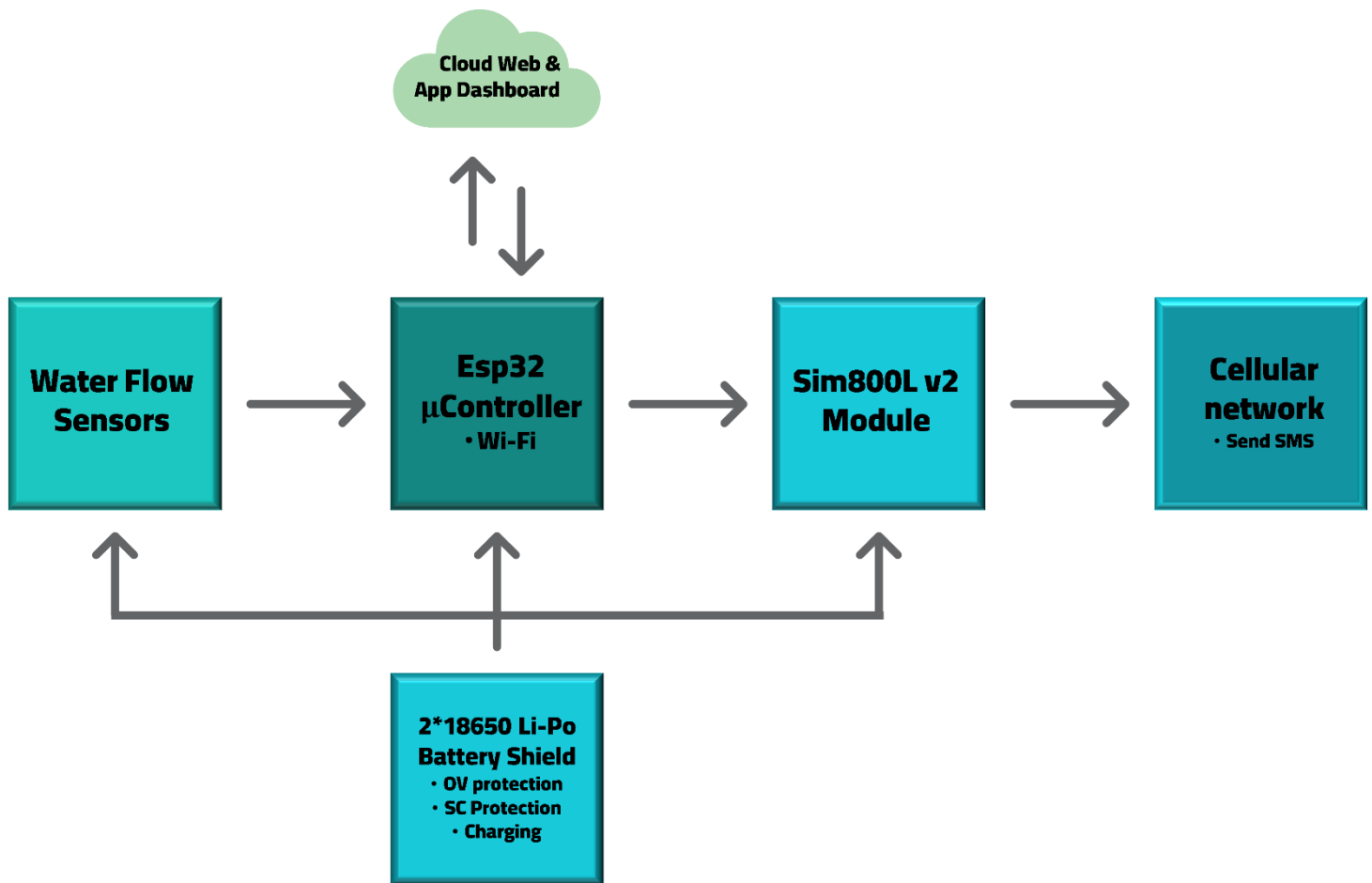
- ESP32 DOIT DevKit V1
- Water Flow Sensor
- Sim800L V2 module
- 18650 Battery shield
- Water Pump
- Water Hose
- plumbing fittings
- Water Tanks

Circuit Diagram



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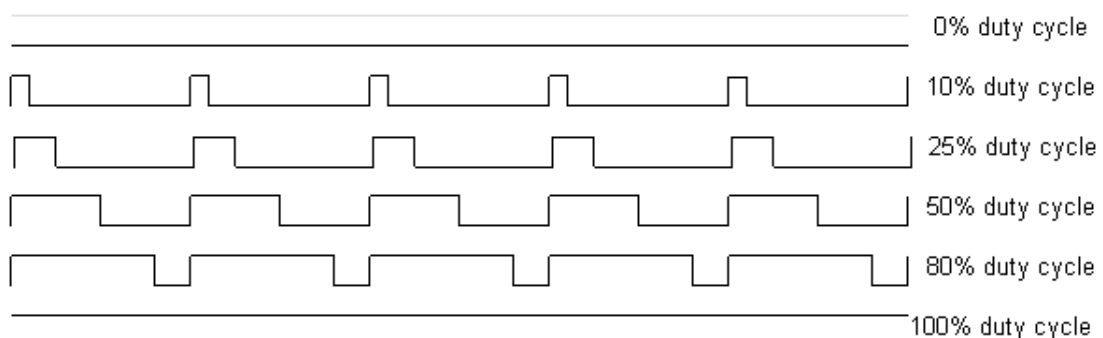
Block Diagram



Water Flow sensor



- **Introduction:** Water flow sensors are devices used to measure the volume of water flowing through a pipe. They are commonly used in water management systems, irrigation systems, and other applications where accurate measurement of water flow is important.
- **Features:** Water flow sensors typically consist of a plastic valve body, a water rotor, and a hall-effect sensor. When water flows through the valve, it rotates the rotor, which changes the speed of the motor. This change is measured by the hall-effect sensor and output as a pulse signal, allowing the rate of flow of water to be calculated.
- **Applications:** Water flow sensors can be used in a variety of applications, including industrial plants, commercial and residential buildings, and public water supply systems. They can be used to monitor the amount of water being supplied and used, and to ensure that water is being used efficiently.
- **Working Principle:** The main components of a water flow sensor are the hall-effect sensor, turbine wheel, and magnet. The water flows in through the inlet and out through the outlet, driving the turbine wheel to turn. The magnet on the wheel turns with it, triggering the hall-effect sensor to output high and low level square waves (pulse) **PWM**.
- **PWM:** Pulse Width Modulation (PWM) is a technique for controlling analog circuits using a digital output from a microcontroller. It works by rapidly switching the output on and off, varying the amount of time the output is on (the pulse width) to control the average voltage or current delivered to the load.
- In the case of a water flow sensor, the hall-effect sensor outputs a PWM signal with a frequency proportional to the flow rate of the water. The duty cycle of the PWM signal, which is the ratio of the pulse width to the period of the signal, represents the flow rate as a percentage of the maximum flow rate that the sensor can measure.
- By measuring the frequency and duty cycle of the PWM signal from the hall-effect sensor, it is possible to calculate the flow rate of water passing through the sensor.



- **Pinout:** The main components YF - S201 has 3 Pins for power and the output. The table below explains the color, pin type, and function of the pin.

Color	Pin type	Function
Red	Power	To provide power to the module
Black	Ground	Connected to the ground terminal
Yellow	Signal	Analog output from the sensor

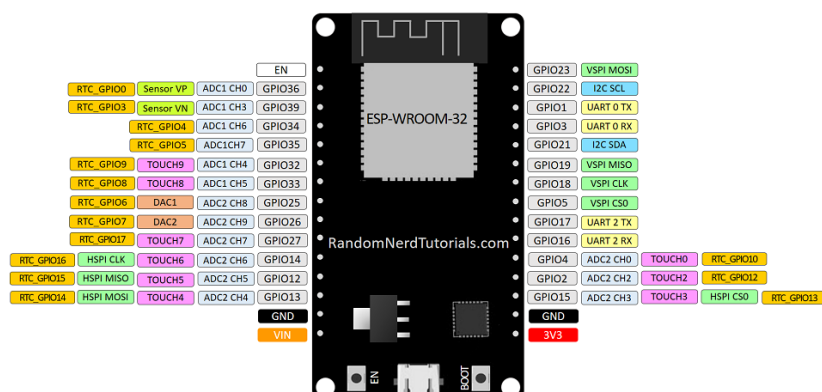


ESP 32 DivKit V1 Microcontroller

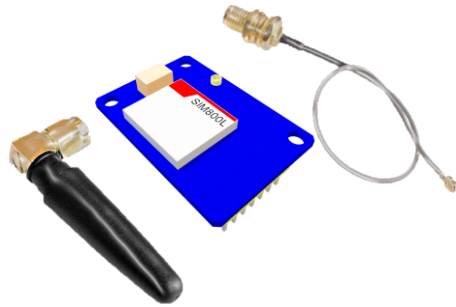


- **Introduction:** The ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth connectivity. It is well-suited for IoT applications, offering a compact and cost-effective solution for connecting sensors and other devices to the cloud.
- **Features:** The ESP32 devkit has 30 pins in total, with most of the I/O pins broken out to the pin headers on both sides for easy interfacing. Developers can either connect peripherals with jumper wires or mount the board on a breadboard. The board can be powered through a Micro-USB port, 3.3V pin, or 5V pin.
- **Applications:** The ESP32 can be used in a wide range of IoT applications, including home automation, environmental monitoring, and remote control. It is commonly used in projects that require wireless connectivity and real-time data processing.
- **Hardware Overview:** The ESP32 devkit is based on the ESP32 microcontroller, which features two CPU cores, 320 KiB of RAM, and 448 KiB of ROM. The board also includes a USB-to-UART bridge for communication with a computer, as well as a range of input/output options.
- **Working Principle:** The ESP32 microcontroller processes data from sensors and other input devices, and transmits this data to the cloud via Wi-Fi. It can also receive data from the cloud and control output devices such as relays or motors. The ESP32 is programmable using the Arduino IDE or other development environments.
- **Pinout Description:** The pinout diagram of the ESP32 devkit shows the location and function of each pin on the board. The pins are labeled according to their function, such as power, ground, input/output, and communication. Some pins have specific functions, such as the Boot-mode and Reset buttons.

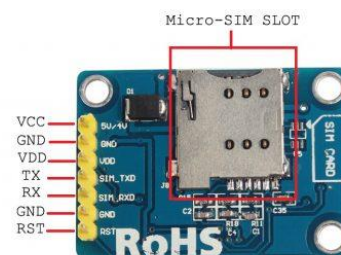
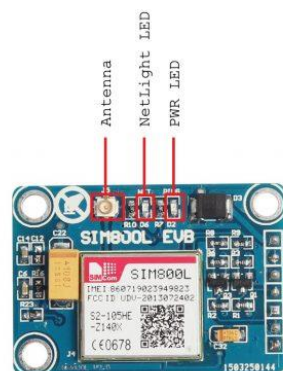
ESP32 DEVKIT V1 – DOIT version with 30 GPIOs



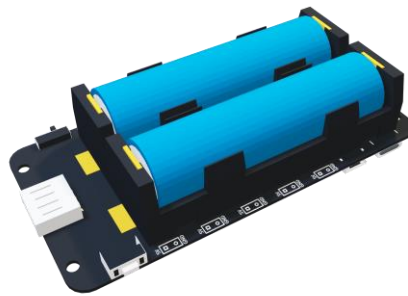
Sim800L v2 (GSM Module)



- **Introduction:** The SIM800L module is a complete Quad-band GSM/GPRS solution in a LGA type which can be embedded in the customer; it has a set of TTL level serial interface, a set of power supply interface. Besides, there are a set of antenna interface on this module.
- **Features:**
 - Operating Voltages: 3.6 – 5.20 Volts
 - Operating Current: 1-2.5 Amps
 - It features Bluetooth, FM and Embedded AT(AT commands)
 - Quad-band 850/900/1800/1900MHz
 - Operation temperature: - 40 ~ 85 degree Celsius
- **Applications:** The Sim800L V2 GSM module can be used in any application that requires SMS notifications or remote control via SMS. It is commonly used in security systems, remote monitoring systems, and IoT projects.
- **Hardware Overview of SIM800L GSM/GPRS module :** This SIM800L in figure II.21, it is a module supports quad-band GSM/GPRS network, available for GPRS and SMS message data remote transmission. SIM800L is one of the most commonly used GSM module among hobbyists and Arduino community. It is not very easy for a beginner to properly understand and use PIC16f628A with SIM800L .
- **Pin out Description:** There are 7 total pins on the SIM800L, which we are going to use to interface with Microcontroller :
 - VCC: External Supply Voltage input for SIM800L
 - GND: External Ground for SIM800L
 - VDD: Microcontroller Supply voltage input for SIM800L
 - RST: Reset pin for SIM800L
 - RXD: Serial communication (Receiver Pin)
 - TXD: Serial communication (Transfer Pin)



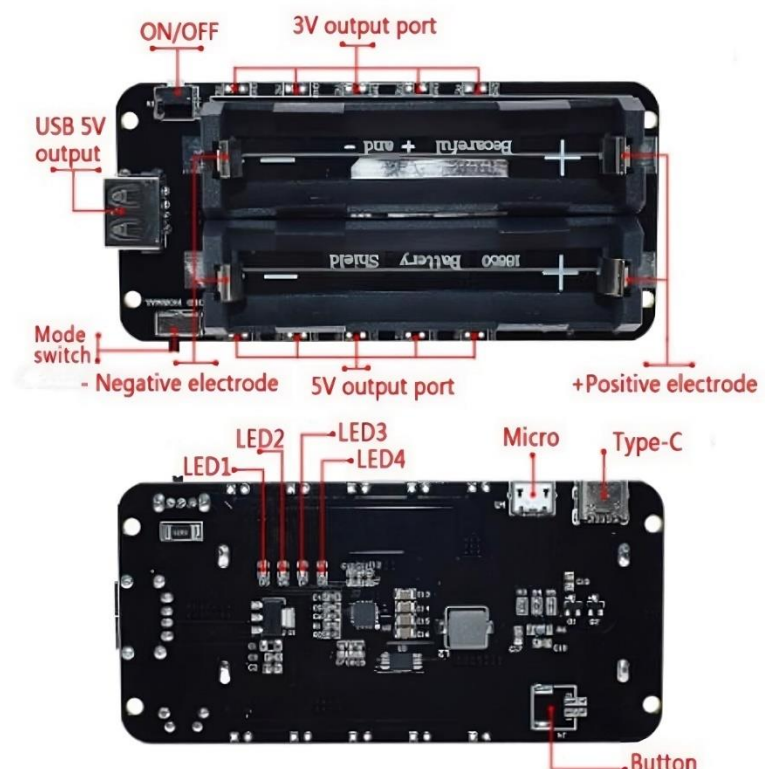
18650 Battery Shield



- **Introduction:** The 18650 battery shield is a device that allows you to power your electronics projects using one or two 18650 lithium-ion batteries. It features a charging circuit, battery protection, and a range of output options.
- **Features:** The 2*18650 battery shield includes a battery holder for two 18650 Li-Ion batteries, a Micro USB and a USB C port for battery charging input, and a range of output options. It provides 5VDC output via a USB type A port, as well as 5x 5VDC (up to 4A) connectors and 5x 3VDC (up to 1A) connectors. The shield also includes battery charging and battery charged LEDs, and a battery protection mechanism for overcharge and deep discharge.
- **Applications:** The 18650 battery shield can be used in any project that requires a portable power source. It is commonly used in wireless projects, where it provides a convenient and reliable power source for microcontrollers, sensors, and other devices.
- **Hardware Overview:** The 2*18650 battery shield consists of three main circuits: a lithium-ion battery charger circuit, a battery protection circuit, and a dc-dc boost converter circuit. The dc-dc boost converter chip is used to deliver 5VDC through the USB type A port and 5V male-header points, generated from the available 3.7V-4.2VDC Li-Ion battery output. The second channel output, marked as 3V but actually 3.3V, is supplied by the parallel combination of three positive voltage regulator chips.

- **Pin out Description:**

- Micro-USB charging interface
- USB-C charging interface
- switch to control USB output
- 5* 3V/1A output
- 5* 5V/2A output
- USB A 5V Female output



Water Pump



This pump will be used to circulate water through the system. It will be controlled separately by a switch, not by the microcontroller.

Water Hose



This hose will be used to connect the various components of the system, allowing water to flow between them.

Plumbing Fittings



These fittings will be used to connect the water hose to the water flow sensor, water pump, and water tanks.

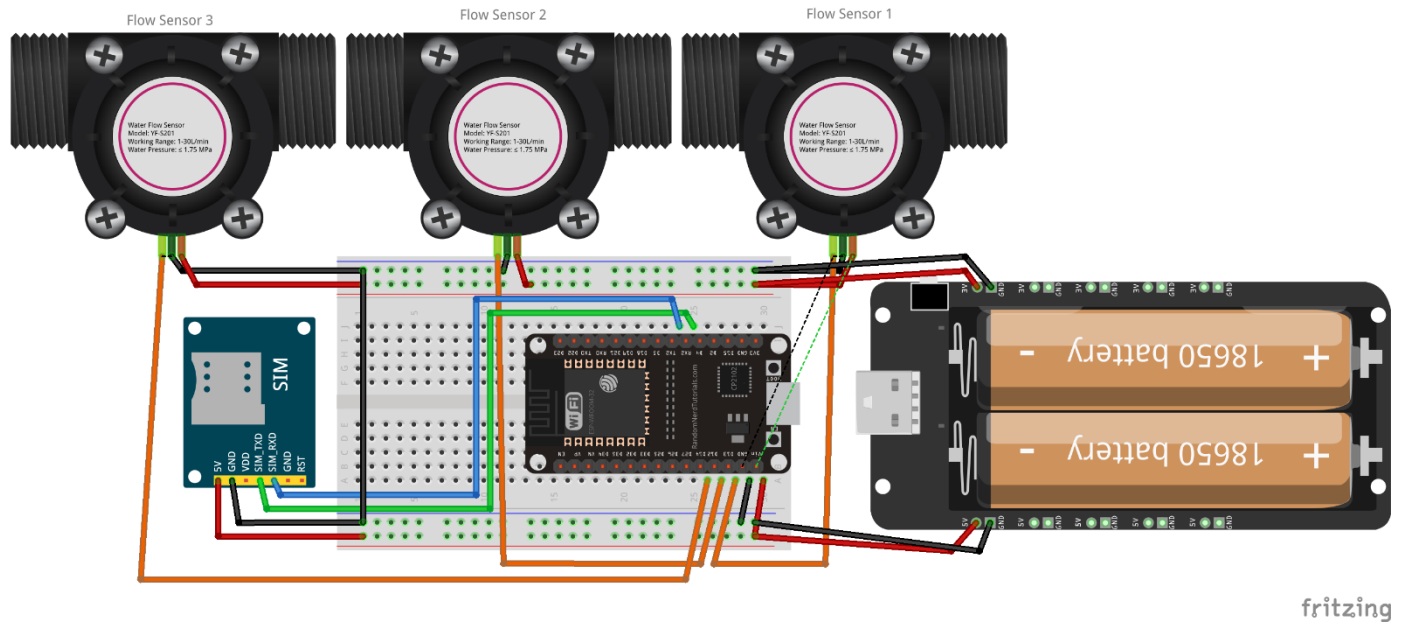
Water Tanks



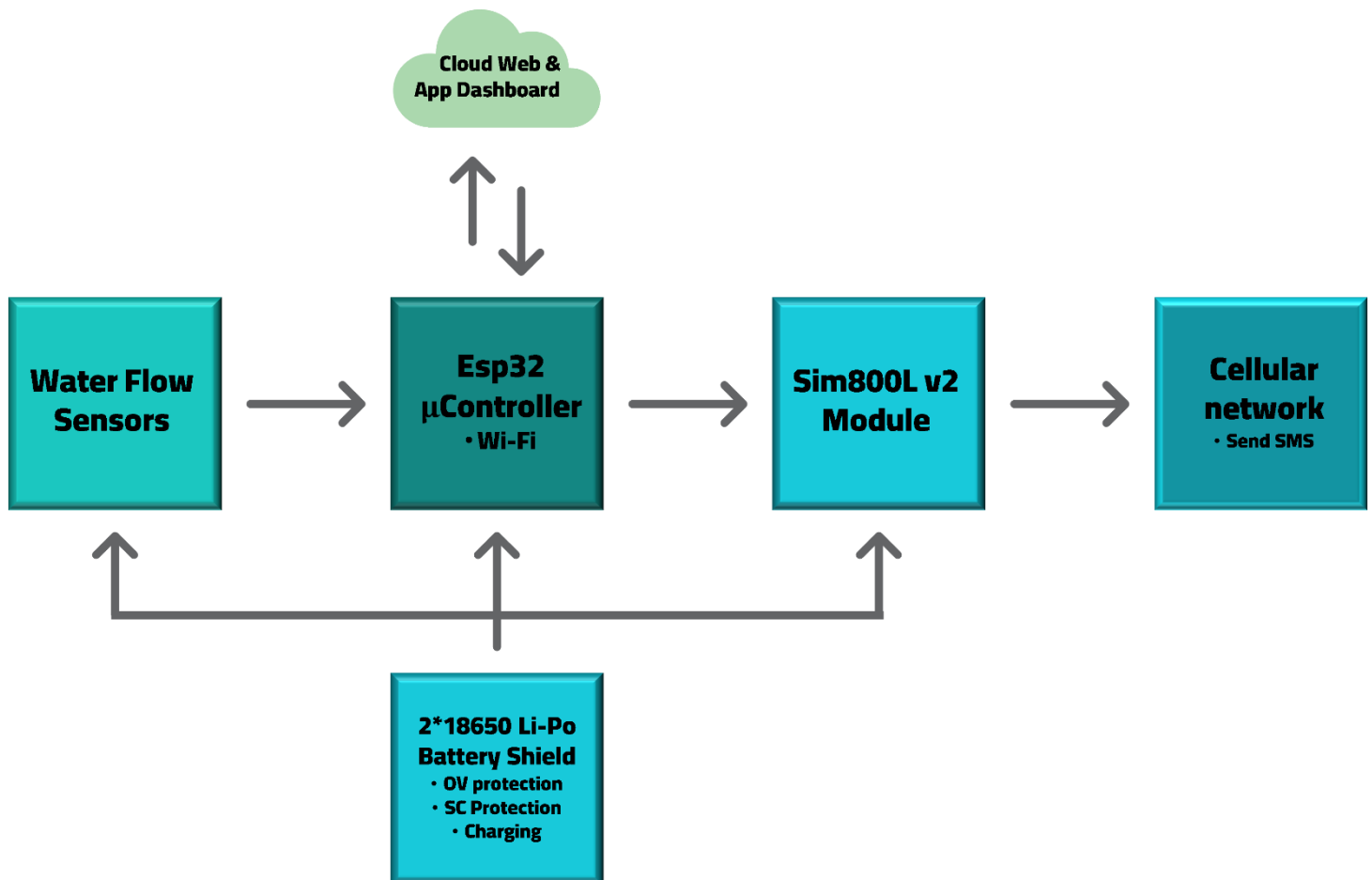
These tanks will store water for use in the system. They will be connected to the water pump and water flow sensor using plumbing fittings and water hose.

Chapter Two

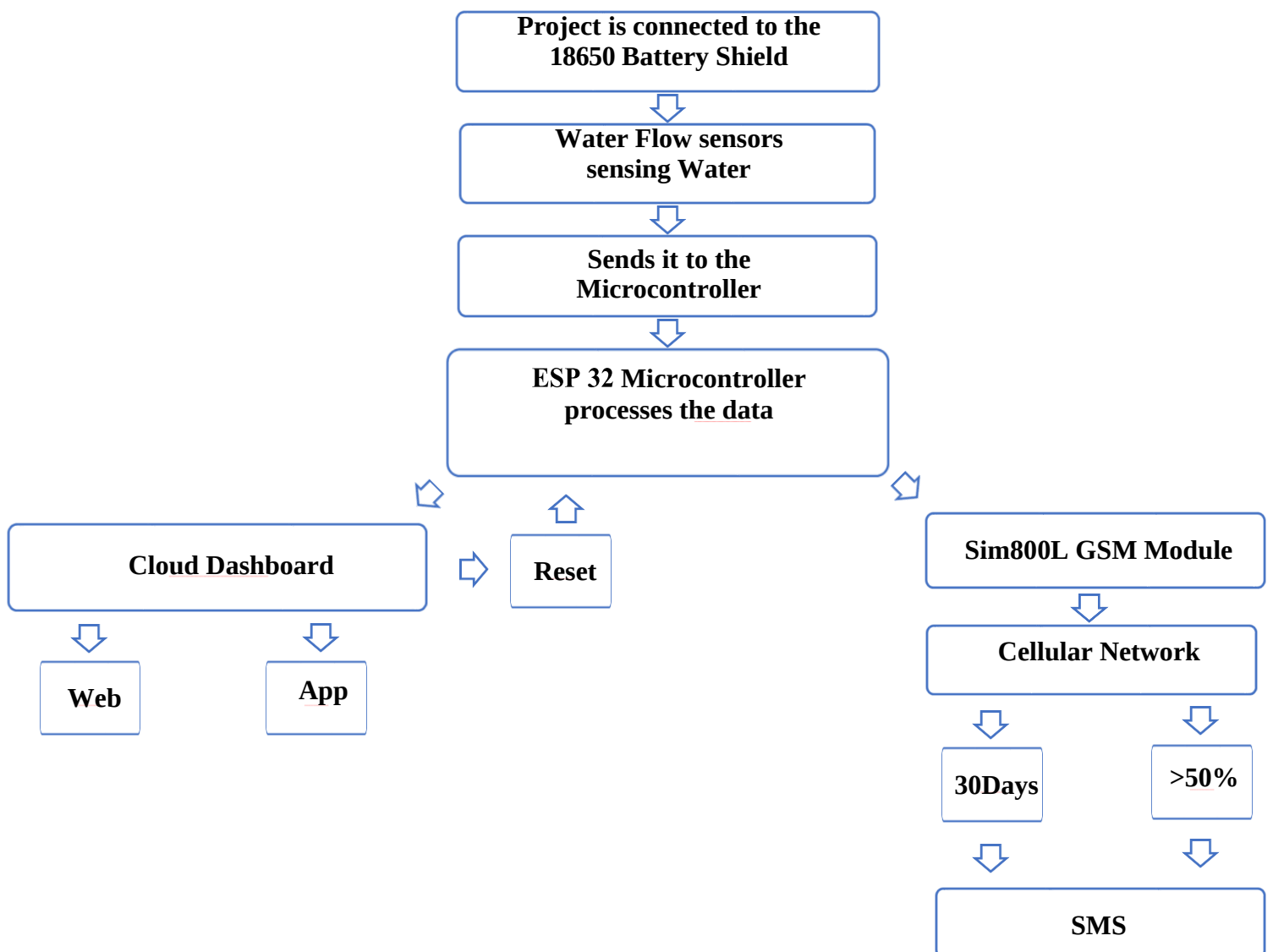
Circuit Diagram



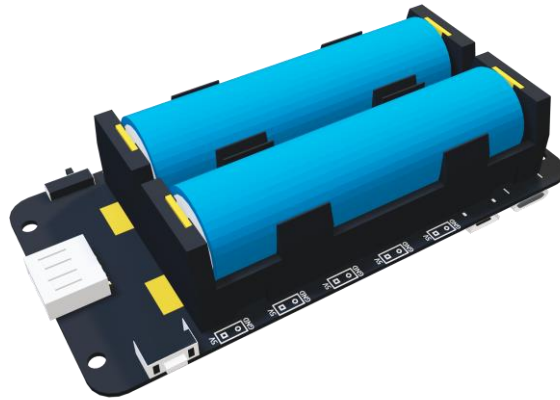
Block Diagram



FLOW CHART



18650 Battery Shield



The 18650 battery shield provides power to the ESP32 DOIT DevKit V1 microcontroller and other components. It may also include features such as charging circuitry, voltage regulation, and protection against overcharging or short circuits.

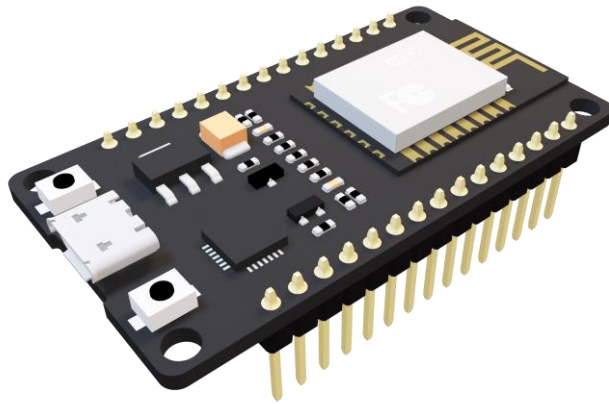
Water Flow sensor



Water flows through the water flow sensor, which measures the volume of water flowing through the pipe.

The water flow sensor sends data to the ESP32 DOIT Devkit V1 microcontroller.

ESP 32 Devkit V1 Microcontroller



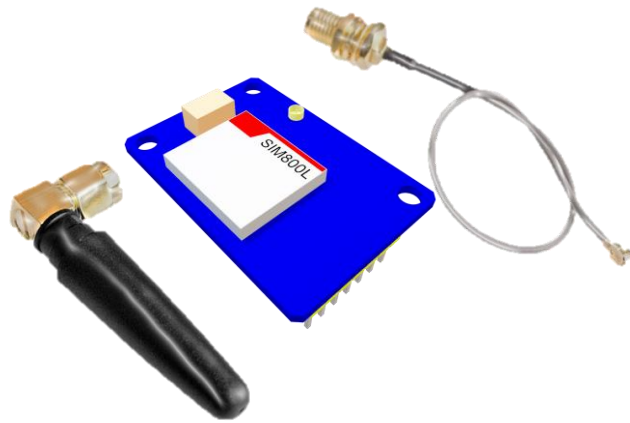
The microcontroller processes the data and sends it to the cloud-based web and app dashboard via Wi-Fi.

Cloud Web & App Dashboard



The dashboard displays real-time data on water usage for property owners and tenants to monitor. It also includes a virtual reset button that allows property owners and tenants to reset the water usage data.

Sim800L v2 (GSM Module)



The Sim800L V2 module sends two types of SMS notifications: one to tenants when they reach 50% of their usage, and another to the property owner every 30 days with the totals for each tenant.

Property owners and tenants can use the dashboard to monitor their water usage and make informed decisions about their consumption.

Chapter Three

Project Result

- The results of this project met the goals we set for ourselves. Our goal was to develop a smart water management system for apartments using IoT technology.
- This project has the potential to improve water metering, reduce conflicts between property owners and tenants, and promote water conservation.
- By providing real-time monitoring of water usage, accurate billing, and promoting water conservation, the system has the potential to deliver significant benefits to both property owners and tenants.

Conclusion

In conclusion, this graduation project aimed to develop a smart water management system for apartments using IoT technology. The proposed system consists of several components, including water flow sensors, a Wi-Fi capable microcontroller, a GSM module, and a cloud-based web and app dashboard. The main objective of the project was to provide a more efficient and cost-effective solution for water metering in apartments.

The results of the project show that the proposed system has the potential to improve water metering, reduce conflicts between property owners and tenants, and promote water conservation. By providing real-time monitoring of water usage, accurate billing, and promoting water conservation, the system has the potential to deliver significant benefits to both property owners and tenants.

Overall, the proposed IoT-based water metering system offers a promising solution for improving water management in apartments. By leveraging the power of IoT technology, the system can provide real-time data on water usage, allowing property owners and tenants to make informed decisions about their consumption. This has the potential to improve the accuracy and transparency of water metering, reducing the potential for disputes and conflicts.

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[esp32_datsheet_en.pdf \(espressif.com\)](#)

[ESP32 Dev board Pinout, Specifications, datasheet and Schematic \(etechnophiles.com\)](#)

[SMS Controller by Using SIM800L V2 : 5 Steps - Instructables](#)

[Sim800L Library Revised - Arduino Reference](#)

[esp32projects/code for Water flow sensor with blynk 2.0 IOT project.ino at main · humam206/esp32projects ·](#)

[GitHub](#)

Landing Page For the Project :



Appendix

Code:

Here is the program we used:

```
#define BLYNK_PRINT Serial // Define a macro that redirects Blynk's debug output to the
Serial interface
#include <WiFi.h> // Include the WiFi library
#include <WiFiClient.h> // Include the WiFiClient library
#include <BlynkSimpleEsp32.h> // Include the Blynk library for the ESP32
char auth[] = "AuthTokenofBlynk"; // Define a character array that stores the Blynk
authentication token
char ssid[] = "MyWifi"; // Define a character array that stores the name of the WiFi network to
connect to
char pass[] = "Pass123123"; // Define a character array that stores the password for the WiFi
network
const int input1 = 13; // Define a constant integer variable that stores the pin number for FS1
const int input2 = 12; // Define a constant integer variable that stores the pin number for FS2
const int input3 = 14; // Define a constant integer variable that stores the pin number for FS3
int X1, X2, X3; // Define integer variables for storing pulse duration values for each of the
three inputs
int Y1, Y2, Y3; // Define integer variables for storing pulse duration values for each of the
three inputs
double TIME1 = 0, TIME2 = 0, TIME3 = 0; // Define double precision floating point variables
for storing time values for each of the three inputs and initialize them to 0
double FREQUENCY1 = 0, FREQUENCY2 = 0, FREQUENCY3 = 0; // Define double
precision floating point variables for storing frequency values for each of the three inputs and
initialize them to 0
double WATER1 = 0, WATER2 = 0, WATER3 = 0; // Define double precision floating point
variables for storing water flow rate values for each of the three inputs and initialize them to 0
double TOTAL1 = 0, TOTAL2 = 0, TOTAL3 = 0; // Define double precision floating point
variables for storing total water usage values for each of the three inputs and initialize them to
0
double LS1 = 0, LS2 = 0, LS3 = 0; // Define double precision floating point variables for
storing liters per second values for each of the three inputs and initialize them to 0
double totalSum = 0; // Define a double precision floating point variable for storing the total
sum of water usage and initializes it to 0
bool alertTriggered = false; // Define a boolean variable that tracks if an alert has been
triggered and initializes it to false
unsigned long startTime; // Define an unsigned long variable that stores the start time of the
program in milliseconds since boot
unsigned long elapsedTime; // Define an unsigned long variable that stores the elapsed time
since startTime in milliseconds
bool isActivated = false; // Define a boolean variable that tracks if a function has been
activated and initializes it to false

void updateSerial() { // Function that updates serial communication between two serial ports
```

```

    delay(500);
    while (Serial.available()) {
        Serial2.write(Serial.read()); // Forward what Serial received to Software Serial Port
    }
    while (Serial2.available()) {
        Serial.write(Serial2.read()); // Forward what Software Serial received to Serial Port
    }
}

void setup() { // Function called once when your program starts
    Serial2.begin(9600);
    delay(3000);
    startTime = millis();
    Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
    Serial.begin(9600);
    pinMode(input1, INPUT);
    pinMode(input2, INPUT);
    pinMode(input3, INPUT);
    Blynk.virtualWrite(V4, 0); // Set the initial state of the button to OFF
    Blynk.virtualWrite(V4, HIGH); // Enable the button to send messages to the board
}

void printLiters(const char* label, double value) { // Function that prints out a label followed
by a value in liters
    Serial.print(label);
    Serial.print(": ");
    Serial.print(value);
    Serial.println(" liters");
    if (value > 0.5 && !alertTriggered) {
        alertTriggered = true; // Set the flag to true to indicate alert has been triggered
        Serial2.begin(9600);
        send_SMS();
    }
}

void send_SMS() { // Function that sends an SMS message using AT commands
    Serial2.println("AT+CMGF=1"); // Configuring TEXT mode
    updateSerial();
    Serial2.println("AT+CMGS=\"" + 9665555555555555 "\""); // Change ZZ with country code and
xxxxxxxxxxxx with phone number to sms
    updateSerial();
    Serial2.print("ALERT: Exceeded 50% of the bill cut"); // Text content
    updateSerial();
    Serial.println();
    Serial.println("Message Sent");
    Serial2.write(26);
}

```

```
}
```

```
void loop() { // Function called repeatedly while your program is running
  updateSerial();
  // Flow sensor 1 PWM calculations
  X1 = pulseIn(input1, HIGH);
  Y1 = pulseIn(input1, LOW);
  TIME1 = X1 + Y1;
  FREQUENCY1 = 1000000 / TIME1;
  WATER1 = FREQUENCY1 / 7.5;
  LS1 = WATER1 / 60;
  if (FREQUENCY1 >= 0) {
    if (isinf(FREQUENCY1)) {
      Blynk.virtualWrite(V1, TOTAL1);
    } else {
      TOTAL1 = TOTAL1 + LS1;
      Blynk.virtualWrite(V1, TOTAL1);
    }
  }
  // Flow sensor 2 PWM calculations
  X2 = pulseIn(input2, HIGH);
  Y2 = pulseIn(input2, LOW);
  TIME2 = X2 + Y2;
  FREQUENCY2 = 1000000 / TIME2;
  WATER2 = FREQUENCY2 / 7.5;
  LS2 = WATER2 / 60;
  if (FREQUENCY2 >= 0) {
    if (isinf(FREQUENCY2)) {
      Blynk.virtualWrite(V2, TOTAL2);
    } else {
      TOTAL2 = TOTAL2 + LS2;
      Blynk.virtualWrite(V2, TOTAL2);
    }
  }
  // Flow sensor 3 PWM calculations
  X3 = pulseIn(input3, HIGH);
  Y3 = pulseIn(input3, LOW);
  TIME3 = X3 + Y3;
  FREQUENCY3 = 1000000 / TIME3;
  WATER3 = FREQUENCY3 / 7.5;
  LS3 = WATER3 / 60;
  if (FREQUENCY3 >= 0) {
    if (isinf(FREQUENCY3)) {
      Blynk.virtualWrite(V3, TOTAL3);
    } else {
      TOTAL3 = TOTAL3 + LS3;
```

```

        Blynk.virtualWrite(V3, TOTAL3);
    }
}
elapsedTime = millis() - startTime; // Calculate the elapsed time
if (!isActivated && elapsedTime >= 30UL * 24UL * 60UL * 60UL * 1000UL) { // If
elapsedTime is greater than or equal to 1 month to send the SMS
    activateFunction(); // Call your desired function
    isActivated = true; // Set the flag to indicate activation
}
}

```

```

void activateFunction() { // Function that sends an SMS message using AT commands
Serial2.println("AT+CMGF=1"); // Configuring TEXT mode
updateSerial();
Serial2.println("AT+CMGS=\"+96655555555555\"");
updateSerial();
Serial2.print("for Owner: Apartment 1 = " + String(TOTAL1) + ",Apartment 2 = " +
String(TOTAL2) + ", Apartment 3 = " + String(TOTAL3));
updateSerial();
}

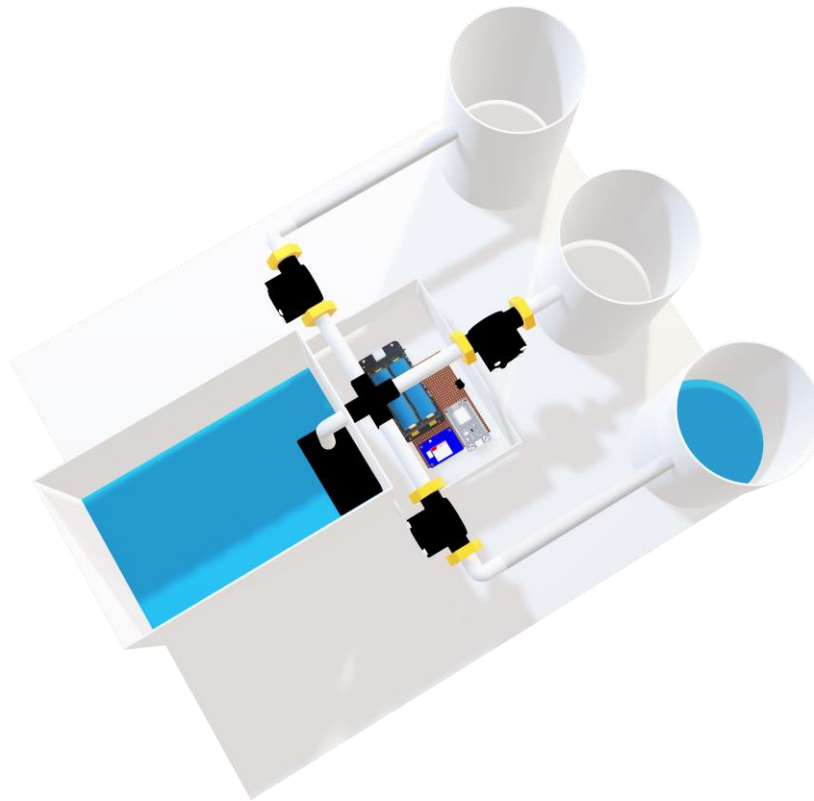
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```

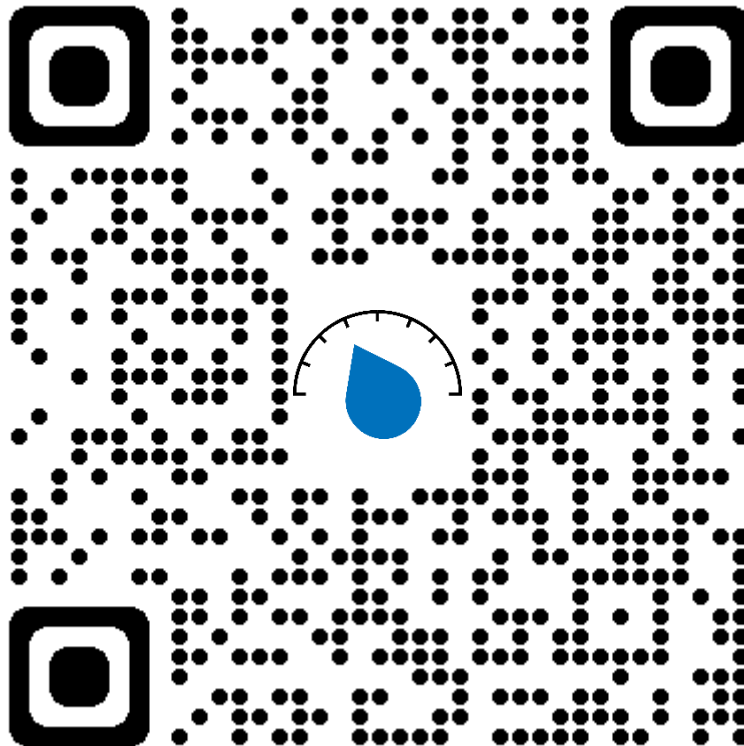
BLYNK_WRITE(V4) { // Function called when virtual pin V4 is updated on the Blynk server
if (param.asInt()) { // Check if the button is pressed
TIME1 = 0, TIME2 = 0, TIME3 = 0;
FREQUENCY1 = 0, FREQUENCY2 = 0, FREQUENCY3 = 0;
WATER1 = 0, WATER2 = 0, WATER3 = 0;
TOTAL1 = 0, TOTAL2 = 0, TOTAL3 = 0;
LS1 = 0, LS2 = 0, LS3 = 0;
}}

```

3D Model:



To View the 3D model on Browser:



Datasheets:

Water Flow Sensor YF-S201 :

MODEL: YF-S201

Description:

Water flow sensor consists of a plastic valve body, a water rotor, and a hall-effect sensor. When water flows through the rotor, rotor rolls. Its speed changes with different rate of flow. The hall-effect sensor outputs the corresponding pulse signal. This one is suitable to detect flow in water dispenser or coffee machine. We have a comprehensive line of water flow sensors in different diameters. Check them out to find the one that meets your need most.

Features:

Compact, Easy to Install
High Sealing Performance
High Quality Hall Effect Sensor
RoHS Compliant

Specifications:

Working Voltage: DC 4.5V~24V
Normal Voltage: DC 5V~18V
Max. Working Current: 15mA (DC 5V)
Load capacity: ≤ 10 mA (DC 5V)
Flow Rate Range: 1~30L/min
Load Capacity: ≤ 10 mA (DC 5V)
Operating Temperature: $\leq 80^{\circ}\text{C}$
Liquid Temperature: $\leq 120^{\circ}\text{C}$
Operating Humidity: 35%~90%RH
Allowing Pressure: ≤ 1.75 MPa
Storage Temperature: -25°C ~ $+80^{\circ}\text{C}$
Storage Humidity: 25%~95%RH
Electric strength 1250V/min
Insulation resistance $\geq 100\text{M}\Omega$
External threads: 1/2"
Outer diameter: 20mm
Intake diameter: 9mm
Outlet diameter: 12mm



Application:

Water heaters, credit card machines, water vending machine, flow measurement device!

Cercuit:

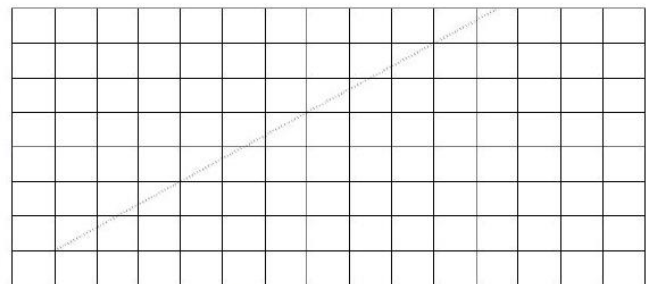
Red: Positive
Black: GND
Yellow: Output signal

Flow Range: 100L/H-1800H-L/H

Flow (L/H)	Frezq.(Hz)	Erro range
120	16	± 10 5%
240	32.5	
360	49.3	
480	65.5	
600	82	
720	90.2	

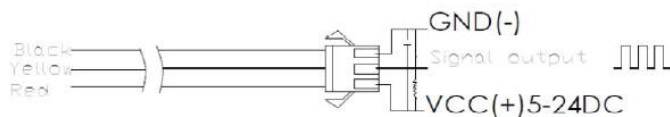
L/H

840
720
600
480
360
240
120
0

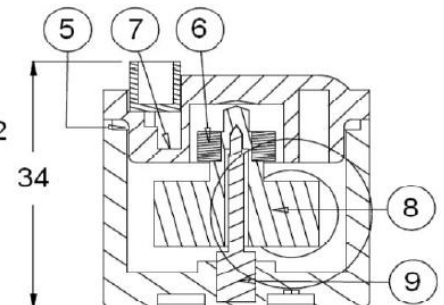
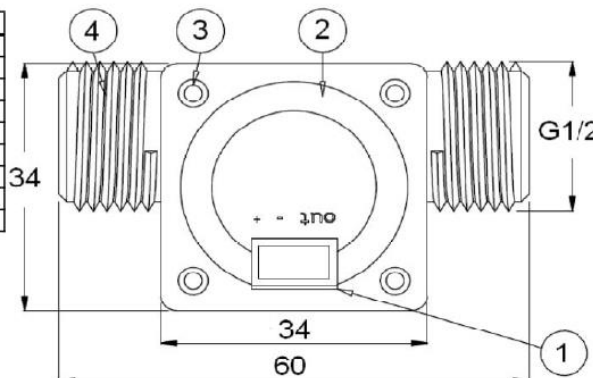


16 32.5 49.5 65.5 82 90.2 Freq.(HZ)

Connection method:



N°	Item	Material
1	Wire	PVC
2	Bonnet	PA
3	Screw	Zinc Plated
4	Valve Body	PA
5	Press Valve	
6	Magnet	
7	Hall	
8	Impeller	POM
9	Steel Sharft	SUS304



Closed

1 Overview

ESP32 is a single 2.4 GHz Wi-Fi-and-Bluetooth combo chip designed with the TSMC low-power 40 nm technology. It is designed to achieve the best power and RF performance, showing robustness, versatility and reliability in a wide variety of applications and power scenarios.

The ESP32 series of chips includes ESP32-D0WD-V3, ESP32-D0WDR2-V3, ESP32-U4WDH, ESP32-S0WD (NRND), ESP32-D0WDQ6-V3 (NRND), ESP32-D0WD (NRND), and ESP32-D0WDQ6 (NRND), among which,

- ESP32-S0WD (NRND), ESP32-D0WD (NRND), and ESP32-D0WDQ6 (NRND) are based on chip revision v1 or chip revision v1.1.
- ESP32-D0WD-V3, ESP32-D0WDR2-V3, ESP32-U4WDH, and ESP32-D0WDQ6-V3 (NRND) are based on chip revision v3.0 or chip revision v3.1.

For details on part numbers and ordering information, please refer to Section 7. For details on chip revisions, please refer to [ESP32 Chip Revision v3.0 User Guide](#) and [ESP32 Series SoC Errata](#).

1.1 Featured Solutions

1.1.1 Ultra-Low-Power Solution

ESP32 is designed for mobile, wearable electronics, and Internet-of-Things (IoT) applications. It features all the state-of-the-art characteristics of low-power chips, including fine-grained clock gating, multiple power modes, and dynamic power scaling. For instance, in a low-power IoT sensor hub application scenario, ESP32 is woken up periodically only when a specified condition is detected. Low-duty cycle is used to minimize the amount of energy that the chip expends. The output of the power amplifier is also adjustable, thus contributing to an optimal trade-off between communication range, data rate and power consumption.

Note:

For more information, refer to Section 3.7 *RTC and Low-Power Management*.

1.1.2 Complete Integration Solution

ESP32 is a highly-integrated solution for Wi-Fi-and-Bluetooth IoT applications, with around 20 external components. ESP32 integrates an antenna switch, RF balun, power amplifier, low-noise receive amplifier, filters, and power management modules. As such, the entire solution occupies minimal Printed Circuit Board (PCB) area.

ESP32 uses CMOS for single-chip fully-integrated radio and baseband, while also integrating advanced calibration circuitries that allow the solution to remove external circuit imperfections or adjust to changes in external conditions. As such, the mass production of ESP32 solutions does not require expensive and specialized Wi-Fi testing equipment.

1.2 Wi-Fi Key Features

- 802.11b/g/n

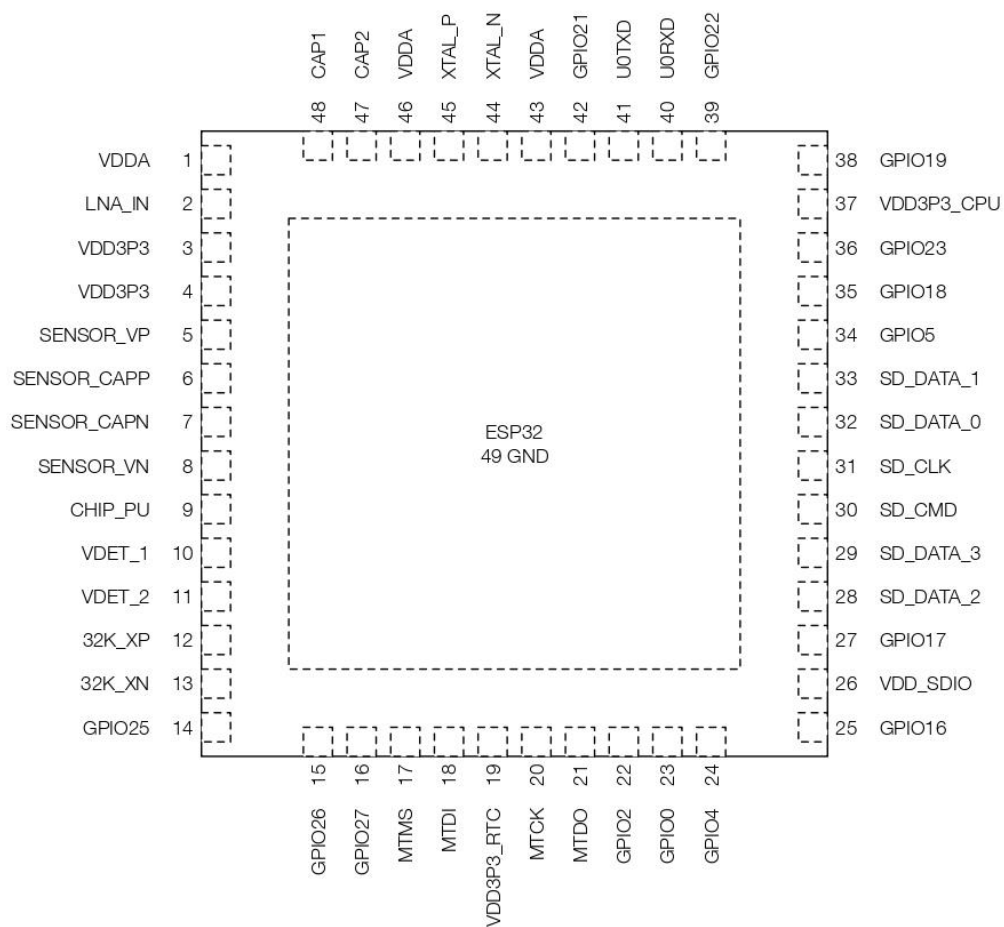


Figure 3: ESP32 Pin Layout (QFN 5*5, Top View)

Note:

For details on ESP32's part numbers and the corresponding packaging, please refer to Section 7 *Part Number and Ordering Information*.



1. Introduction

This document describes SIM800H&SIM800L hardware interface in great detail. The document can help customer to quickly understand SIM800H&SIM800L interface specifications, electrical and mechanical details. With the help of this document and other SIM800H&SIM800L application notes, customer guide, customers can use SIM800H&SIM800L to design various applications quickly.

2. SIM800H&SIM800L Overview

SIM800H&SIM800L is a quad-band GSM/GPRS module, that works on frequencies GSM850MHz, EGSM900MHz, DCS1800MHz and PCS1900MHz. SIM800H&SIM800L features GPRS multi-slot class 12/class 10 (optional) and supports the GPRS coding schemes CS-1, CS-2, CS-3 and CS-4.

With a tiny configuration of 17.8*15.8*2.4mm, SIM800H&SIM800L can meet almost all the space requirements in customer applications, such as smart phone, PDA and other mobile devices.

SIM800H&SIM800L is a LGA package with 88 pads, and provides all hardware interfaces between the module and customers' boards.

- Support 5*5*2 keypads
- One full modem serial port, customer can configure two serial ports
- One USB, the USB interfaces can debug, download software
- Audio channel which includes two microphone input; a receiver output and a speaker output
- Programmable general purpose input and output.
- A SIM card interface
- Support Bluetooth(only SIM800H)
- Support FM
- Support one PWM

SIM800H&SIM800L is designed with power saving technique so that the current consumption is as low as 1.04mA in sleep mode.

2.1. SIM800H&SIM800L

Table 1: Module information

	SIM800H	SIM800L
GSM	850,900,1800 and 1900MHz	850,900,1800 and 1900MHz
BT	Support	Not support
FLASH	32Mbit	16Mbit
RAM	32Mbit	32Mbit

2.2. SIM800H&SIM800L Key Features

Table 2: SIM800H&SIM800L key features

Feature	Implementation
Power supply	3.4V ~4.4V
Power saving	Typical power consumption in sleep mode is 1.04mA (BS_PA_MFRMS=9)

3. Package Information

3.1. Pin Out Diagram

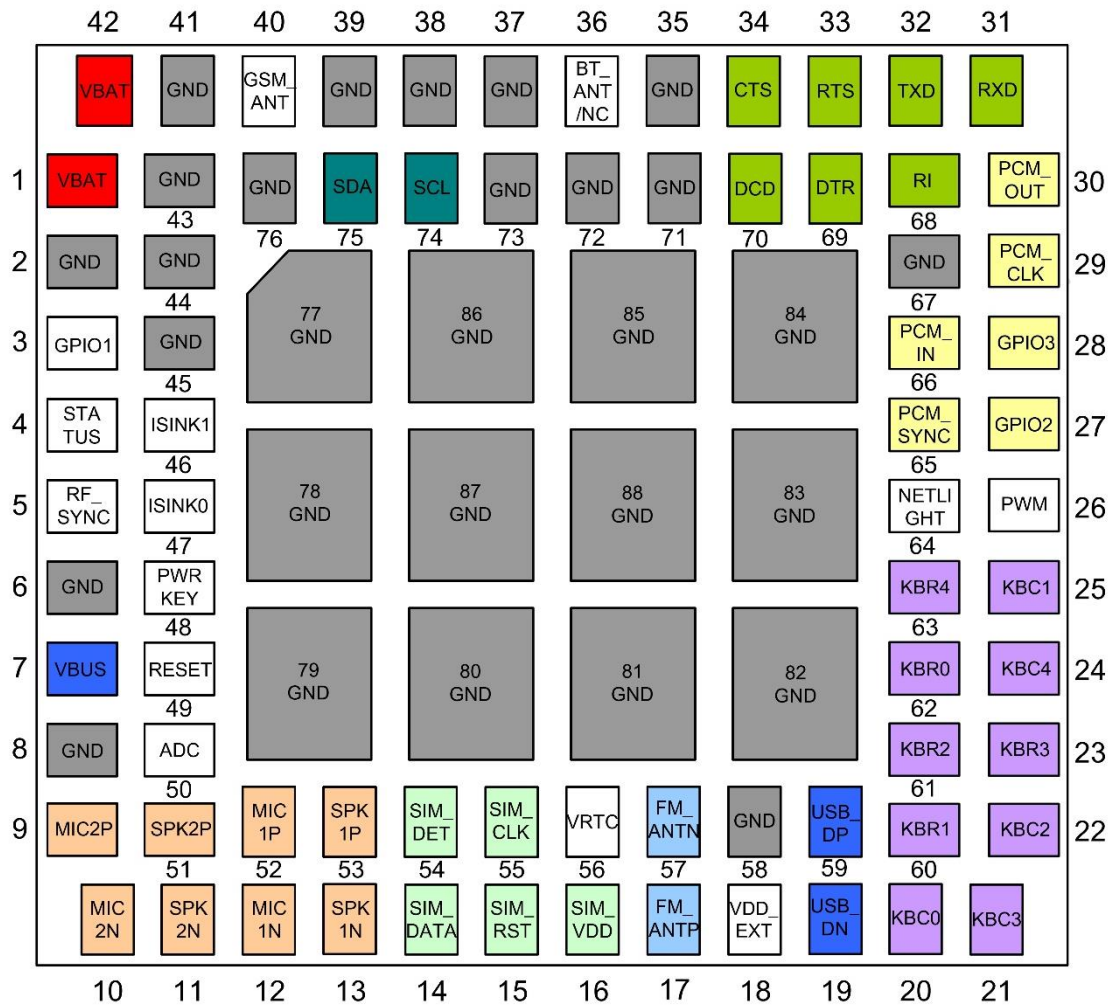


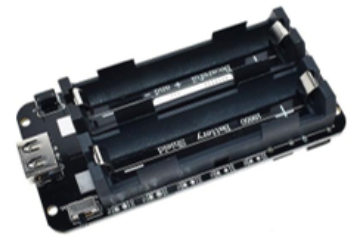
Figure 2: Pin assignment (Top view)

2*18650 battery shield:

Model: 2*18650 Battery Shield

Description:

The 18650-battery shield is a device that allows you to power your electronics projects using one or two 18650 lithium-ion batteries. It features a charging circuit, battery protection, and a range of output options

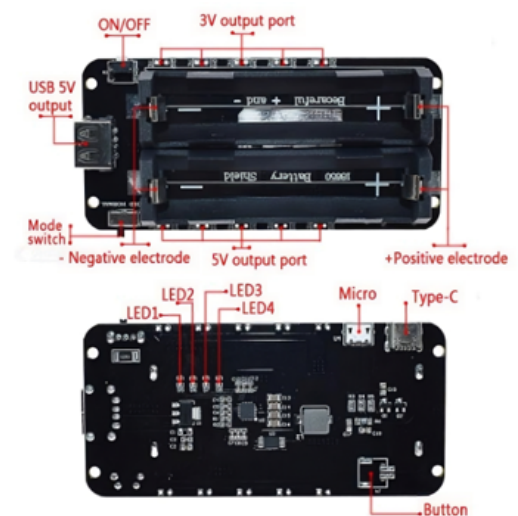


Features:

Input overvoltage protection, output overcurrent/short circuit protection, charging timeout/overvoltage protection

Specifications:

with a Micro-USB charging interface
with USB-C charging interface
Charging current: max. 0.5A
with a switch to control USB output
Input voltage: 5~8V
with 5* 3V/1A output
with 5* 5V/2A output
with status light (red = under charge, green = full charge)
Dimensions: 9.8×4.8cm x 2.1



Applications:

The 18650-battery shield can be used in any project that requires a portable power source. It is commonly used in wireless projects, where it provides a convenient and reliable power source for microcontrollers, sensors, and other devices.

Circuit:

The background electronics are a clever blend of three requisite circuits, that is to say, a lithium-ion battery charger circuitry, a battery protection circuitry, and a dc-dc boost converter circuitry.

In the shield, the dc-dc boost converter chip is used to deliver 5VDC (actually 4.99V) through the USB 'A' socket and 5V male-header points, generated from the available 3.7V-4.2VDC Li-Ion battery output. The second channel output, marked as 3V but 3.3V on the dot, is supplied by the parallel combination of three positive voltage regulator chips. The 3.3VDC supply is derived from the 3.7V-4.2VDC Li-Ion battery output (not from the 5VDC output)

Inputs:

USB-C and micro-USB Outputs: 5x 5V outputs, 5x 3.3V outputs, and a 5V USB-A output

First Weekly Report

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الكلية التقنية بالرياض
قسم الهندسة الإلكترونية التطبيقية

Course Name: Project (Bachelor) Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.1)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:30PM	Riyadh College of Technology	2023 / 03 / 19	Sunday	442139580	Abdullah Alshenaifi
				439233863	Abdullah Aldosari
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
- View the previous week's report. - Discussing the additions of the project. - Sending the weekly report.	- A project proposal after being agreed upon by the group members. - Work on new additions after presenting them to supervisors. - Sending the weekly report to the supervisors after working on it by the group and explaining what has been done after obtaining approval for the project. Prepare to work on the next step to find pieces for the project. - Prepare to work on the next step to find pieces for the project.

Supervisor signature

Trainee signature

Second Weekly Report

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Course Name: Project (Bachelor)Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.2)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:30PM	Riyadh College of Technology	2023 / 04 / 03	Sunday	442139580	Abdullah Alshenaifi
				439233863	Abdullah Aldosari
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
- Viewing and discussing final additions to the project. - The agreement and final preparation of the entire report for the project. - Approval from supervisors to start the project.	- View the project report and discuss additions. - Accepting suggestions and additions from supervisors, if any. - Add some graphic modifications to the project summary to clarify further.

Supervisor signature

Trainee signature

Third Weekly Report

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Course Name: Project (Bachelor)Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.3)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:30PM	Riyadh College of Technology	2023 / 04 / 09	Sunday	442139580	Abdullah Alshenaifi
				439233863	Abdullah Aldosari
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
- Find the elements for the project. - Visiting shops and pricing parts.	Colleagues Mohammed Haddadi Abdullah Aldosari visited electronics stores to find out their availability and to take a price quote, where some pieces were found and some were requested so that we could start the project.

Supervisor signature

Trainee signature

Fourth Weekly Report

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Course Name: Project (Bachelor) Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.4)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:45PM	Riyadh College of Technology	2023 / 04 / 30	Sunday	442139580	Abdullah Alshenaifi
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
Collect items from electronic shops.	- Ordering remaining items online (Water Flow Sensor). - Visiting some specialized stores for boxes and equipment, so that we can install and assemble electronic circuit in them.

Supervisor signature

Trainee signature

Fifth Weekly Report

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Course Name: Project (Bachelor)Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.5)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:30PM	Riyadh College of Technology	2023 / 05 / 07	Sunday	442139580	Abdullah Alshenaifi
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
Start working on project.	- Preparing box for electronic circuit to be ready to install circuit as soon as it is ready. - Receive items ordered online. - Work on arrangement of circuit and how to connect it.

Supervisor signature

Trainee signature

Sixth Weekly Report

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Course Name: Project (Bachelor) Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.6)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:45PM	Riyadh College of Technology	2023 / 05 / 14	Sunday	442139580	Abdullah Alshenaifi
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
Work is in progress on electronic circuit by connecting it.	Connect some electronic boards to each other.

Supervisor signature

Trainee signature

Seventh Weekly Report

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Course Name: Project (Bachelor) Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.7)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
02:15PM	Riyadh College of Technology	2023 / 05 / 23	Tuesday	442139580	Abdullah Alshenaifi
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
Completing project by adding some final touches.	- Programming is finished and has been added to project. - Working on some improvements to circuit. - Circuit is working well, trying to add on/off switch to circuit to save power.

Supervisor signature

Trainee signature

Eighth Weekly Report

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Course Name: Project (Bachelor)Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.8)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:30PM	Riyadh College of Technology	2023 / 06 / 04	Sunday	442139580	Abdullah Alshenaifi
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
- Some error replacement. - Continuing writing final report and PowerPoint presentation.	- We encountered several problems with our code, which required us to spend additional time debugging and troubleshooting. - We decided to switch from using an ESP8266 microcontroller to an ESP32 microcontroller, which required us to make changes to our code and hardware setup.

Supervisor signature

Trainee signature

Nineth Weekly Report

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Course Name: Project (Bachelor)Course Code: (4252 Elk) Accredited units: 2 Weekly Communication Units: 4 Office Hours: third training semester for the year (1444 AH)	Name of trainer: Abdulaziz bin Abdullah Alrodhyan Office No .: 13 F 3B Email: aalrodhyan@tvtc.gov.sa Phone: +966555204404
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Periodic meetings of the project material (meeting no.9)

Time	Meeting location	Date of the meeting	Day	No.academic	trainer's name
12:45PM	Riyadh College of Technology	2023 / 06 / 11	Sunday	442139580	Abdullah Alshenaifi
				441225780	Mohammed Haddadi
				442104508	Mohammed Alshaya

Tasks agreed at the next meeting to be achieved by the next meeting (points)	Summary of issues discussed (points)
- Some error replacement. - Continuing writing final report and PowerPoint presentation.	- We continued working on debugging our code and making improvements. - We encountered some water leakages in our system, which required us to make additional repairs and adjustments.

Supervisor signature

Trainee signature